

SANGYOON BAE

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Scientific ML • Foundation Models • Physics-Informed AI • Causal Discovery

Spatiotemporal Learning • Representation Learning • Self-Supervised Learning • Distribution Shift

I build foundation models for complex, non-stationary temporal data by embedding the structure of the data-generating process—physics, acquisition geometry, signal statistics—directly into model architecture and training. My methods target causal discovery under indirect observation, multi-scale temporal modeling with state-space architectures, and self-supervised pretraining at scale. Validated across neuroimaging, electrophysiology, and climate time-series; extending toward broader non-stationary decision environments.

Research Impact: 2 sole first-author papers (ICLR 2026, Communications Biology), 2 sole first-author papers under review, and 1 co-first-author paper (ICASSP 2024); 6+ peer-reviewed publications, with models evaluated on datasets up to 49k+ subjects across neural, climate, and population-scale data.

EDUCATION

Ph.D. Candidate, Artificial Intelligence

Sep 2021 – Present

Interdisciplinary Program in AI, Seoul National University, Seoul, South Korea

B.S. in Biological Sciences

Mar 2015 – Aug 2021

Seoul National University, Seoul, South Korea

Student-Designed Minor: Computational Neuroscience for Human Behavior and Cognition. Wet-lab training: neurosurgical procedures (cranial window implantation, brain tissue sectioning), forward genetic screening in *C. elegans*, PCR, Western blot.

SELECTED RESEARCH PROJECTS

INCAMA | Causal Discovery from Non-Stationary Indirect Measurements via Physics-Aware Inversion

(under review)

- Formalized causal discovery from indirect neuroimaging as a conditional identifiability problem: physics-aware inversion (HRF deconvolution, EEG source localization) coupled with delay-aware Mamba-based latent causal discovery exploiting nonstationarity as identifying variation. Proved identifiability reduction with inversion-error propagation bound.
- 2–3× F1 improvement over observation-space and two-stage baselines on TVB simulations (EEG and fMRI); zero-shot transfer to HCP motor-task fMRI (n=1,079) recovers canonical visuo-motor pathways with 4× higher recall than strongest baseline. ([arXiv:2602.09034](#))

PIMSM | Physics-Informed Multi-Scale Mamba for Stable Representations under Distribution Shift

(under review)

- Formalizes temporal kernel mismatch as a failure mode for scientific foundation models under distribution shift. PIMSM derives scale-specific SSM discretization parameters from spectrum-estimated knee frequencies, anchored in acquisition time units, with A-scale regularization to prevent compensatory rescaling.
- Highest representation stability (CKA 0.812) under severe temporal truncation on HCP fMRI (~1k subjects, 360 ROIs); strongest low-resource accuracy at 1% supervision; lowest variable-wise MAE across all horizons and variables on Weather-5K held-out-station spatial OOD forecasting (15.2% average MAE reduction vs. best baseline). ([arXiv:2605.16351](#))

POYO-CAP | Curriculum Pretraining Resolving Scaling Collapse in Heterogeneous Scientific Data

ICLR

2026

- Identified a scaling failure in SSL pretraining on heterogeneous recordings: indiscriminate training causes “scaling collapse” where adding sensors degrades performance. Solution: use higher-order statistics (skewness/kurtosis) to identify statistically regular subsets and adopt a data-informed curriculum—pretrain on predictable signals first, then fine-tune on stochastic ones.
- Unlocks monotonic scaling and 1.98× data efficiency. Validated on 200k+ recordings. ([link](#))

MBBN | Frequency-Band Attention for Multi-Scale Spatiotemporal Foundation Modeling *Communications Biology*

- Derives attention mechanism from the physics of scale-free power-law spectra: decompose signals into domain-informed frequency bands, apply specialized self-attention per band. Physics-grounded pretraining loss (network communicability) anchors optimization.
- Trained at scale on 49,673 individuals; SOTA classification with up to 41.36% AUROC improvement. ([link](#))

RESEARCH EXPERIENCE

Visiting Researcher, Mila – Quebec AI Institute, Montreal 2024

Foundation models for scientific data, advised by Prof. Blake Richards

- Collaborated on scalable foundation models for heterogeneous scientific data. Developed the data-centric curriculum insight—statistical regularity as a pretraining criterion—that became the core contribution of POYO-CAP (ICLR 2026).

Visiting Researcher, Brookhaven National Laboratory 2023

Quantum machine learning infrastructure, advised by Dr. Shinjae Yoo

- Collaborated on quantum machine learning infrastructure; applied privacy-preserving techniques to federated learning settings. Resulted in ICASSP 2024: “Quantum Privacy Aggregation of Teacher Ensembles (QPATE).”

Visiting Researcher, Institut de Neurosciences des Systèmes, Aix-Marseille Université 2022

Biophysical simulator-based modeling, advised by Prof. Viktor Jirsa

- Worked with Prof. Viktor Jirsa on adapting The Virtual Brain (TVB) biophysical simulation framework; experience with simulator-based forward models directly seeded the inverse-problem formulation of the INCAMA project.

Undergraduate Intern, Laboratory of Neurobiology, Seoul National University 2018

Advised by Prof. Bong-Kiun Kaang

- Assisted neurobiology experiments investigating neural circuits in mouse models, including neurosurgical procedures (cranial window implantation) and brain tissue preparation for microscopy.

Undergraduate Intern, Laboratory of Genes and Development, Seoul National University 2017

Advised by Prof. Junho Lee

- Forward genetic screening in *Caenorhabditis elegans* to identify genes regulating dauer developmental pathways.

PUBLICATIONS

Preprints

- **Bae, S., Yoo, S., Cha, J.** (2026) PIMSM: Physics-Informed Multi-Scale Mamba for Stable Neural Representations under Distribution Shift. *arXiv:2605.16351*. ([link](#))

- **Bae, S.**, Oprescu, M., Park, D., Yoo, S., Cha, J. (2026) Latent-Space Causal Discovery from Indirect Neuroimaging Observations. *arXiv:2602.09034*. ([link](#))
- Kim, S., Kwon, J., Kwon, J., **Bae, S.**, Lin, Y., Yoo, S., & Cha, J. (2024). Macro2Micro: Cross-modal Magnetic Resonance Imaging Synthesis Leveraging Multi-scale Brain Structures. *arXiv:2412.11277*. ([link](#))

Peer-Reviewed

- **Bae, S.**, Azabou, M., Richards, B., Cha, J. (2026) Decoding Dynamic Visual Experience from Calcium Imaging via Cell-Pattern-Aware Pretraining. *ICLR 2026*. ([link](#))
- **Bae, S.**, Kwon, J., Cha, J., Yoo, S. (2026). Learning brain dynamics across distinct scaling regimes reveals psychiatric signatures. *Communications Biology*. ([link](#))
- Park, J., **Bae, S.**, Seo, J., Chen, S., Tseng, H., Cha, J. & Yoo, S. (2025) Resting-state fMRI Analysis using Quantum Time-series Transformer. *IEEE QCE 2025*. ([link](#))
- Watkins, W., Wang, H., **Bae, S.**, Tseng, H. H., Cha, J., Chen, S. Y. C., & Yoo, S. (2024). Quantum Privacy Aggregation of Teacher Ensembles (QPATE). *ICASSP 2024*. ([link](#))
- Kim, P., Kwon, J., Joo, S., **Bae, S.**, Lee, D., Jung, Y., ... & Moon, T. (2024). SwiFT: Swin 4D fMRI Transformer. *NeurIPS 2023*. ([link](#))
- Kim, M., Shim, Y., Kwon, J., **Bae, S.**, ... & Kwon, J. S. (2023). Resting-state theta phase-gamma amplitude coupling as a biomarker for the transdiagnostic dimensional approach in psychiatric disorders. *IBRO Neuroscience Reports*, 15. ([link](#))

SCHOLARSHIPS AND AWARDS

- [Best Paper Award, Co-Author] IEEE International Conference on Quantum Computing and Engineering (QCE), 2025
- IPAI Development Fund Scholarship, Seoul National University, 2024
- Sung-Am Educational Foundation Scholarship, South Korea, 2023
- Graduate School of Artificial Intelligence Fellowship, Seoul National University, 2021–2023

TECHNICAL SKILLS

- **Scientific Computing:** Python, NumPy, SciPy, Pandas
- **Machine Learning:** PyTorch (expert), JAX/Flax (XLA/JIT), state-space models (Mamba/SSM), transformers, self-supervised and contrastive pretraining, foundation model training at scale, inverse problem solvers, physics-informed neural networks, causal discovery; agentic AI for accelerated experimentation and analysis
- **Scientific Domains:** heterogeneous, indirectly observed measurements (temporal, spatial, population-scale); time-series and signal processing; multifractal and spectral analysis, dynamical systems, biophysical forward models; demonstrated domains: neuroimaging (fMRI, EEG, calcium imaging), climate time-series (WEATHER-5K)
- **Systems:** multi-GPU training (8× RTX 3090, 4× H100), HPC clusters, CUDA; biophysical simulators (TVB)

MENTORING

Mentored 8 undergraduates, master's students, and high-school students across neuroscience and AI; led weekly reading and code reviews; resulted in peer-reviewed publications.